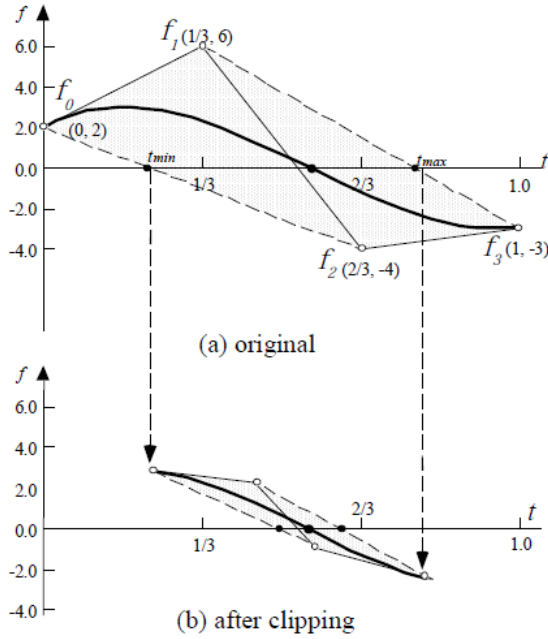


Fig. 1: Solving to polynomial by Bezier clipping Fig. 2: Curve/line intersection.

2.2 Solving to Polynomials

Polynomials can be converted to a Bezier curve. For example, a degree three polynomial can be converted to the cubic Bezier function. Fig. 1(a) shows the polynomial ($f(x) = 24x^3 - 42x^2 + 2x + 2$) converted to the following a cubic Bezier function;

$$f(t) = \sum_{k=0}^3 f_k B_k^3(t) \quad (1)$$

where $(k/3, f_k)$ ($k=0, \dots, 3$) are control points of the Bezier function, and B_k^3 is the Bernstein function. The shaded region is the convex hull of the curve. The root of the curve always exists within the intersection between the axis and the convex hull; that is, the interval $t_{\min}$ and $t_{\max}$. By clipping the curve at $t_{\min}$ and $t_{\max}$, we can get a new curve with thinner convex hull as shown in Fig. 1(b). As the remaining part of the curve approaches a straight line, the intersection interval between the convex hull and the t axis rapidly narrows at the next step. By repeating this process we can find the root.

Iteration terminates when the intersection interval between the convex hull and the t axis is smaller than the given tolerance. In the first step, in the example of Fig. 1, the interval is 0.55, but in the third iteration, the interval is only 0.0003. After three iterations we can find the intersection point.

If the intersection between the convex hull and the axis is relatively large, there is a possibility of multiple roots. In that case, the curve is subdivided at the mid point into two curves. And Bezier clipping can then be applied to each curve.

5.1 Field function

In the metaball technique, a free-form surface is defined as an isosurface (equi-potential surface) of a field function. The field value at any point is defined by distances from the specified points in space. We used the degree six field function proposed by Wyvill[Wyvill86]. If two balls are placed at the same location, it has twice the volume of the isosurface for a single ball. Thus, for geometric modeling, degree six polynomial function is useful expressed by

$$f_i(r) = -\frac{4}{9}\left(\frac{r}{R_i}\right)^6 + \frac{17}{9}\left(\frac{r}{R_i}\right)^4 - \frac{22}{9}\left(\frac{r}{R_i}\right)^2 + 1 \quad (8)$$

where R_i is the radius of metaball i and r is the distance from a point to the center $P_i(x_i, y_i, z_i)$.

For n metaballs, the shape of the curved surface is defined by the points satisfying the following equation.

$$f(x, y, z) = \sum_{i=1}^n q_i f_i - T = 0 \quad (9)$$

where T is a threshold, q_i the density values at the center of metaball i .

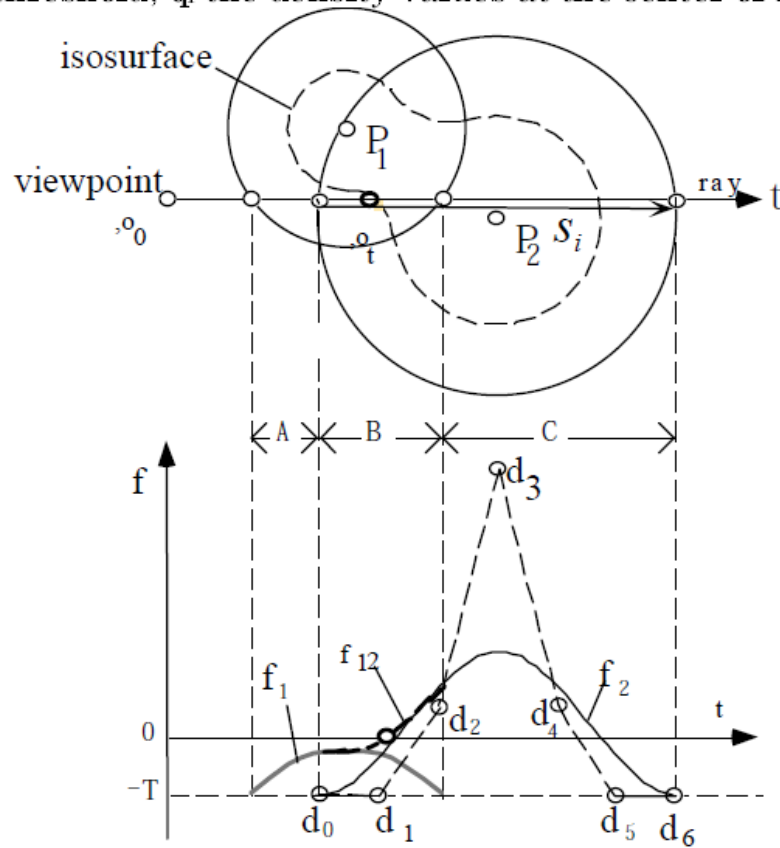


Fig. 6: Density distribution on the ray.

5.2 Intersection Test between Ray and Metaballs

The main task for rendering metaballs is intersection tests between rays and isosurfaces. In our algorithm[Nishi94a], the field function on the ray is expressed by Bezier functions, so the root of this function is effectively and precisely solved by Bezier clipping.

Let's discuss the intersection test between a ray and multiple metaballs. Fig. 6 shows a ray and an isosurface defined by two balls and shows the density distribution on the ray. By using parameter s_i ($0 < s_i < 1$) on the intersected interval with ball i (see Fig.6), the density is expressed by

$$d(s_i) = \sum_{k=0}^6 d_k B_k^6(s_i) \quad (10)$$

where $d_0 = d_1 = d_5 = d_6 = 0$, $d_2 = d_4 = \frac{16}{27}a_i^2$, $d_3 = 8\frac{(8a_i+5)a_i^2}{45}$, and a_i is (intersected length/ R_i)² ($0 < a_i < 1$).

As shown in Fig. 6, Bezier curves, f_1 and f_2 , are clipped by the interval to be tested (i.e., section B in the figure), then both of the clipped curves are composited. This composting of the curves is very simple. It is performed by simply adding each control point d_{ki} belonging to f_1 and f_2 ; After composting the curves, the new curve f_{12} is also expressed by a degree six Bezier curve, then the root (e.g., P_t in Fig. 6) is found by using Bezier Clipping.

For opaque objects, only one intersection point closest to the viewpoint is required. Once the first root (i.e., a minimum real root) is found, the process is stopped. In the examples in section 8, the average number of iterations for a single intersection is only 2.0.